

Introduction to Data Science and Machine Learning

Data Science: Theory, Practice and Societal Implications

ht 2026

Aurora Esteban Toscano

Part I – Orientation and motivation



1. Outline of the course

- DT8063 – Data Science: Theory, Practice and Societal Implications is a 2nd cycle course part of the MSc in Information Technology.
- Mostly taught in English
- 7.5 credits
- In the study period 1: weeks 36-43: September 1 – October 26
- Structure
 - 6 lectures
 - 1 invited talk
 - 8 labs and project sessions
 - 1 Q&A for seminars
 - 1 seminar presentation on you
 - 1 project presentation on you

1. Outline of the course

- Lecturers
 - Aurora Esteban Toscano (aurora.esteban.toscano@hh.se)
 - Lectures 1, 2, 7, seminars
 - Sepideh Pashami (sepideh.pashami@hh.se)
 - Lecture 4, invited talk
 - Canberk Özen (canberk.ozen@hh.se)
 - Labs, final project
 - Ahmad Ghazawneh (ahmad.ghazawneh@hh.se)
 - Lecture 3

1. Outline of the course

- Course Materials
 - Lecture slides will be uploaded to Blackboard
 - Lab descriptions as well as seminar and project instructions will be available in Blackboard in the corresponding assignments
 - Relevant materials to complete the labs will be provided as well (e.g. notebooks, templates)
 - The schema of the course will be updated in TimeEdit (link provided in Blackboard)
 - These materials are not exhaustive: you are responsible of taking the needed notes to understand the lessons, and make the best from the teaching time by asking questions.

2. What will we learn during the course?

- Define data science and machine learning
- Recognize common data-science problems
- How to follow a rigorous scientific approach when solving a data problem
- Understand what is the role of Generative AI in the data science landscape
- Ethics and fairness when working with data
- Digital service innovation
- Sustainability
- Data privacy

3. How the course is assessed

- **No written or final exam**

Examination	Credits	Grading	Main evicence
Data Science Seminar (2502)	2 hp	U/3/4/5	Scientific report/presentation + peer review + defense
Programming Exercises (2503)	3 hp	U (Fail) / G (Pass)	Code/notebooks from lab assignments
LLM Application Project (2501)	2.5 hp	U/3/4/5	Project report + oral presentation + peer review

- Individual work for the programming exercises
- Group work for seminar & project (2 people per group)

Part II – What is Data Science?



4. Data science in the world around us

Entertainment & Recommendations

- Content personalization
- Churn prediction



Finance & Banking

- Fraud detection
- Risk assessment



Healthcare & Medicine

- Early diagnostics
- Personalized treatments



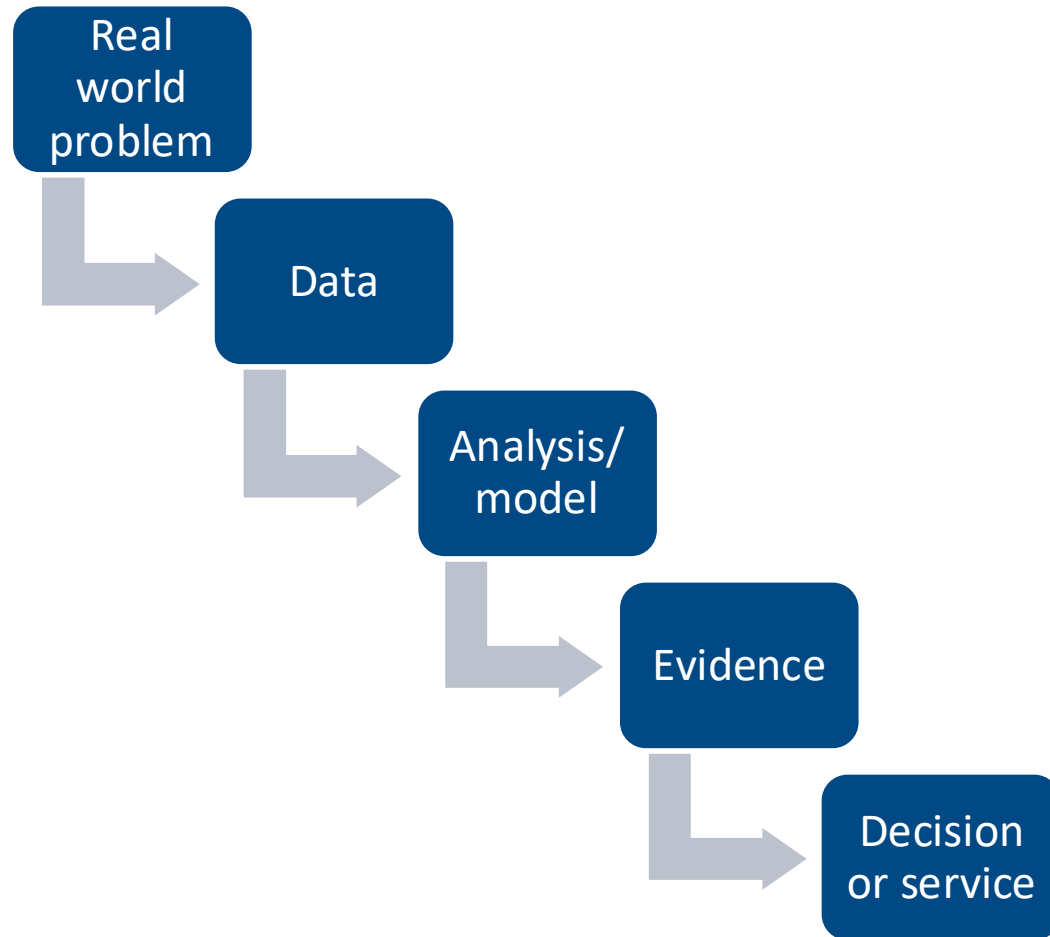
Logistic & Transport

- Route optimization
- Traffic and ETA forecasting



What connect these applications?

5. From data to decisions



Collecting data is not itself data science

6. What is data science?

Data science is an interdisciplinary field that uses math, statistics, computer programming, and business knowledge to extract meaningful insights and actionable knowledge from raw data.

“The goal is to turn data into information, and information into insights”.

Carly Fiorina, 2004

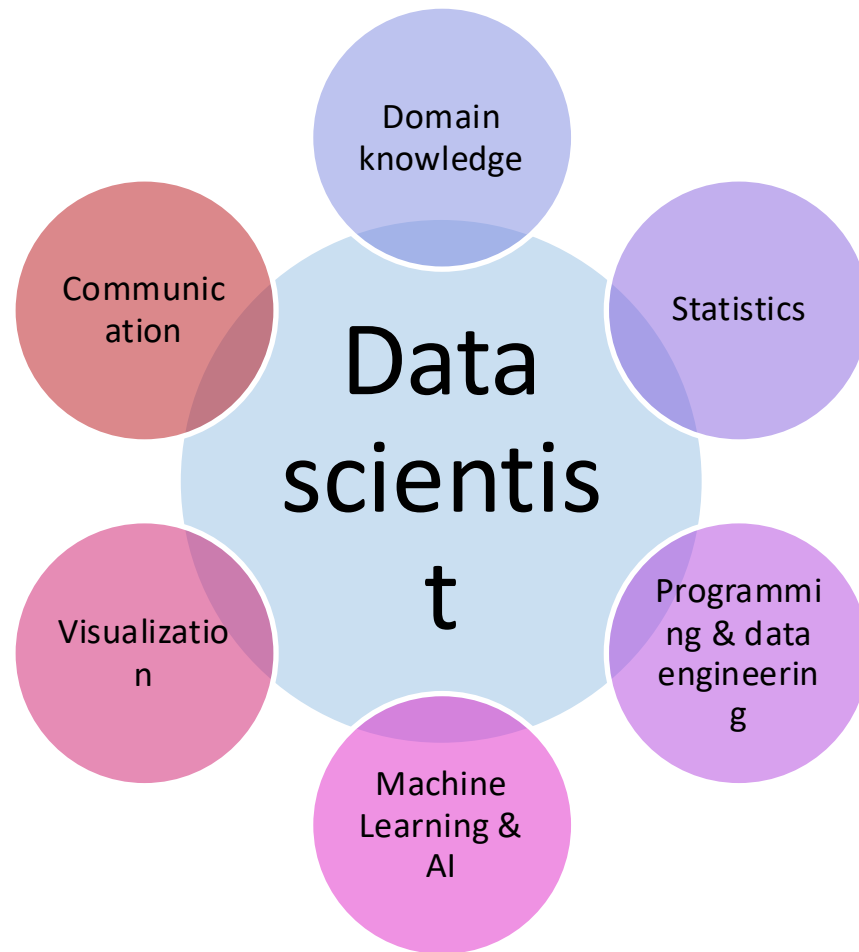
“A field of study that uses scientific methods, processes, and systems to extract knowledge and insights from data.”

The U.S. Census Bureau, 2012

“Data science is only as much of a science as it facilitates the interpretation of data—a two body problem, connecting data to reality. Data alone are hardly a science; regardless how big they get and how skillfully they are manipulated”.

Judea Pearl, 2018

7. The interdisciplinary data scientist



Data science is usually a team activity—but every practitioner needs enough breadth to connect the pieces

Some data scientists are

- Statisticians
- Machine learning experts
- Data miners
- Software engineers
- ...

7. The interdisciplinary data science

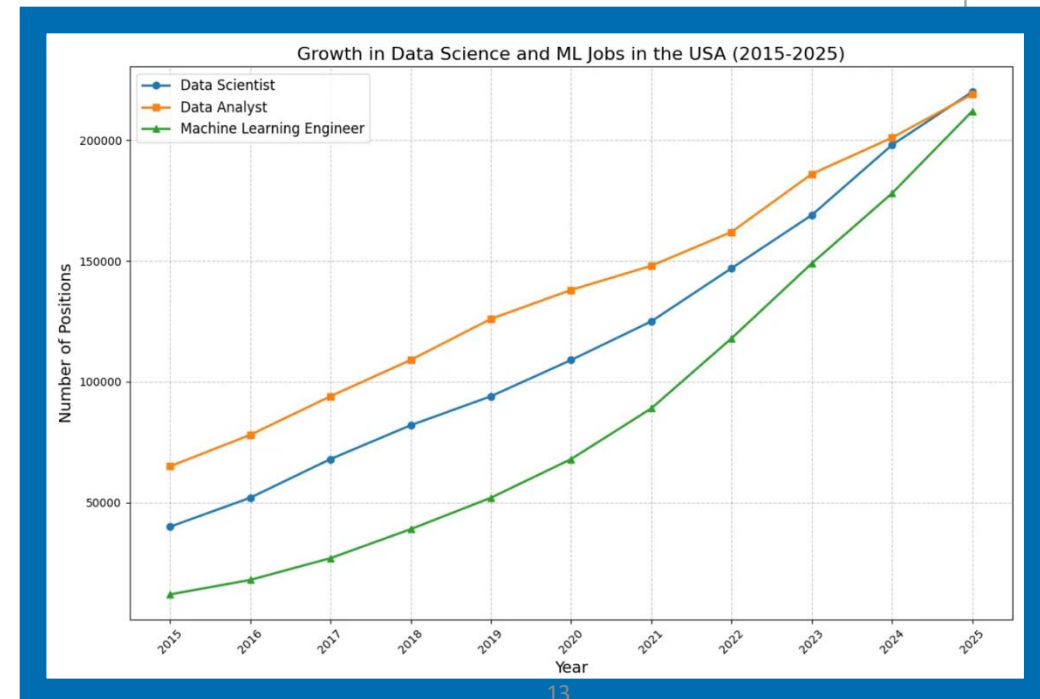


What becomes more valuable

- Framing the right problem
- Understanding the application domain
- Designing credible evaluations
- Identifying bias, risk and limitations
- Integrating models into real systems
- Communicating evidence to decision-makers
- Working effectively across disciplines

The market is moving from one generalist title toward hybrid and specialized roles

- The broad “data scientist” role is divided into data analyst, data engineer, machine-learning engineer and AI/LLM engineer.
- Demand is growing for AI, machine learning, NLP, LLM and production-oriented expertise.
- Continuous learning of technical skills increasingly important.



8. What kinds of questions can data answer?

“Data: The quantities, characters, or symbols on which operations are performed by a computer, which may be stored and transmitted in the form of electrical signals and recorded on magnetic, optical, or mechanical recording media”.

Oxford dictionary

Describe – What happened? • Example: How many patients were readmitted within 30 days last year?	Explain – Why did it happen? • Example: Which factors are associated with patient readmission, and what mechanisms might explain them?
Predict – What may happen? • Example: Which current patients are most likely to be readmitted within 30 days?	Support decisions – What should we do? • Example: Which patients should receive additional follow-up support, given limited resources and possible harms?

Description is not explanation. Correlation is not causation. An accurate prediction does not automatically produce a good decision.

9. Data comes in different forms

- **Structured Data**

- Organized in rows and columns (tables)
- Examples: Financial transactions, inventory records, sensor readings
- Easy to store and query using traditional databases

Emp#	Name	Job	HiringD	Salary	Bonus	Dept#
7369	Red	Engineer	17/12/80	1600.00	500.00	20
7499	Andrews	Technician	20/02/81	800.00	?	30
7521	White	Technician	20/02/81	800.00	100.00	30
7566	Pink	Manager	02/04/81	2975.00	?	20
7654	Martin	Secretary	28/09/81	800.00	?	30
7698	Black	Manager	01/05/81	2850.00	?	30
7782	Neri	Engineer	01/08/81	2450.00	200.00	10
7788	Scott	Secretary	09/11/81	800.00	?	20
7839	Dare	Engineer	17/11/81	2800.00	300.00	10
7844	Turni	Technician	08/09/81	1500.00	?	30
7876	Adams	Engineer	23/09/81	1100.00	500.00	20
					?	30
					?	20
					150.00	10
					?	10

- **Semi-structured Data**

- Has some organizational properties but not fully tabular
- Examples: XML, JSON, server logs, emails
- Flexible format; requires specialized processing

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- Created from PDF via Acrobat SaveAsXML -->
<!-- Mapping Table version: 28-February-2003 -->
<TaggedPDF-doc>
  <?xpacket begin="" id="W5M0MpcEhHzreSzNTczkc9d"?>
    <?xpacket begin="" id="W5M0MpcEhHzreSzNTczkc9d"?>
      <x:xmpmeta xmlns:xmp="http://www.adobe.com/xap/1.0/" rdf:about="">
        <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#">
          <rdf:Description xmlns:xmp="http://ns.adobe.com/xap/1.0/" rdf:about="">
            <xmp:CreateDate>2011-04-06T17:22:05Z</xmp:CreateDate>
            <xmp:CreatorTool>ESRI ArcSOC 9.2.0.1324</xmp:CreatorTool>
            <xmp:ModifyDate>2011-04-07T08:17:15-06:00</xmp:ModifyDate>
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          </rdf:Description>
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            <xmpMM:DocumentID>uuiid:323e04f1-1033-4483-be82-e60a7573f2ec</xmpMM:DocumentID>
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          </rdf:Description>
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            <dc:format>xml</dc:format>
          </rdf:Description>
        </x:xmpmeta>
      <?xpacket end=""?>
    <?xpacket end=""?>
  </Figure>
  <ImageData src="images/WA" data-bbox="664 644 854 684"/>
</TaggedPDF-doc>
```

- **Unstructured Data**

- No predefined format; rich in content
- Examples: Text documents, images, video, audio, social media
- High storage and processing complexity

Google search results for "hadoop big data". The results show approximately 21,400,000 results in 0.71 seconds. The top result is "What is Hadoop? | SAS" with a link to https://www.sas.com/sas/insights/big_data/insights_hadoop.html. Below this, there are "People also ask" questions: "What is the difference between big data and Hadoop?", "How does Hadoop work in big data?", "What is Apache Big Data?", and "What is big data in detail?". Further down, there are links to "Hadoop - Big Data Overview - Tutorialspoint" and "What is Hadoop? - Definition from Whats.com". At the bottom, there is a link to "Welcome to Apache Hadoop" with the URL <http://hadoop.apache.org/>.

A diagram titled "Big Data Hadoop" illustrating the Hadoop ecosystem. It shows a central "Hadoop" node connected to various components: "NameNode", "DataNode", "MapReduce", "Hive", "Pig", "Mahout", "HBase", "Tez", "Oozie", "Ambari", "Cloudera", "Hortonworks", "EMC", "NetScout", "Splunk", "Logstash", "Elasticsearch", "Kafka", "Storm", "Flink", "Spark", "Docker", "Kubernetes", "Ansible", "SaltStack", "Puppet", "Chef", "Vagrant", "VirtualBox", "VMware", "Oracle", "Microsoft", "IBM", "HP", "Dell", "Cisco", "Juniper", "Hewlett-Packard", "Fujitsu", "Hitachi", "Toshiba", "Panasonic", "Sony", "Samsung", "LG", "Huawei", "Zte", "Xiaomi", "Oppo", "Vivo", "Nokia", "Motorola", "Alcatel", "Ericsson", "Nokia", "Huawei", "Zte", "Xiaomi", "Oppo", "Vivo", "Nokia", "Motorola", "Alcatel", "Ericsson". The diagram also includes a section for "Big Data & Hadoop" with a link to https://www.sas.com/sas/insights/big_data/insights_hadoop.html.

10. Scale is only one challenge

Big Data: High-volume, high-velocity, and/or high-variety information that requires innovative, cost-effective processing to enable insights, decision-making, and automation.

The V's of Big Data



Volume

Massive amounts of data generated every second



Velocity

Data flows in at high speed, requiring real-time or near-real-time processing



Variety

Data comes in multiple formats – structured, unstructured, semi-structured



Veracity

The quality, accuracy, and trustworthiness of data



Value

Turning data into meaningful insights and actionable outcomes



Variability

Inconsistent or changing data flows that require adaptive processing

Additional challenges:

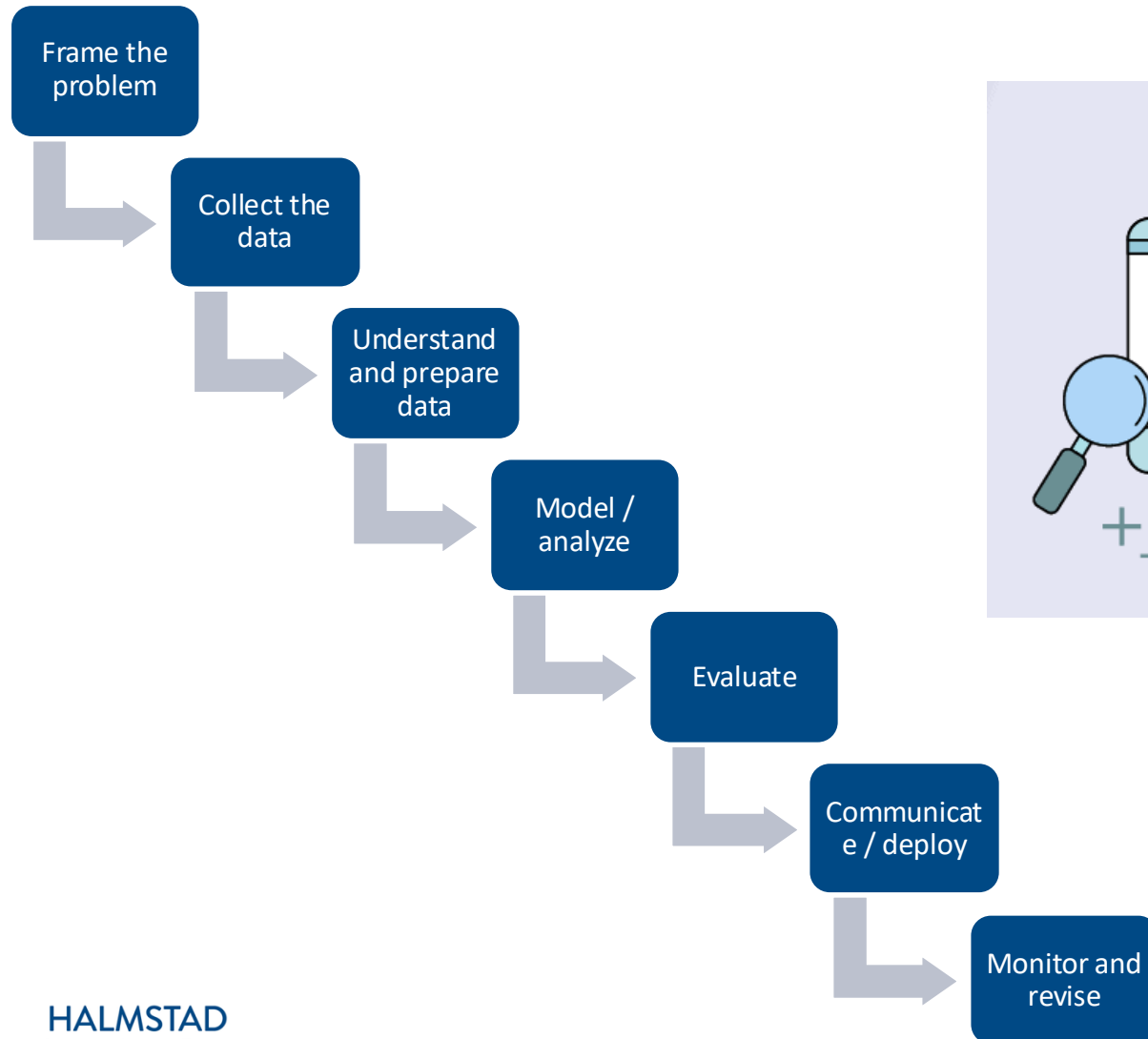
- Quality
- Privacy
- Representativeness
- Information storage and analytics

More data does not automatically mean better evidence

Part III – The Data Science Workflow



11. The data-science workflow



12. Start with the problem, not the algorithm

A useful data-science project begins with a real need—not with a preferred technology.

Before analyzing data, ask:

- **Who** experiences the problem?
- **What** outcome should improve?
- **Which decision or action** could the analysis support?
- **What constraints** must be respected?
- **What would success look like?**

Two ways of creating a problem statement:

“Build a machine-learning model for hospital readmission”

“Help the discharge team identify patients who may need additional follow-up, with the aim of reducing avoidable 30-day readmissions without overwhelming clinical staff”

13. From question to measurable task

Real-world problem:

Which patients may need additional support after leaving hospital?

Element	Measurable definition
Unit of analysis	One hospital discharge
Prediction moment	Immediately before discharge
Available inputs	Diagnoses, previous admissions, length of stay and information available at discharge
Outcome	Unplanned readmission within 30 days: yes or no
Analytical task	Estimate the probability of readmission for each patient
Intended use	Help staff prioritize follow-up support
Success criterion	Identify a useful proportion of high-risk patients without generating too many unnecessary interventions

14. Understand the data

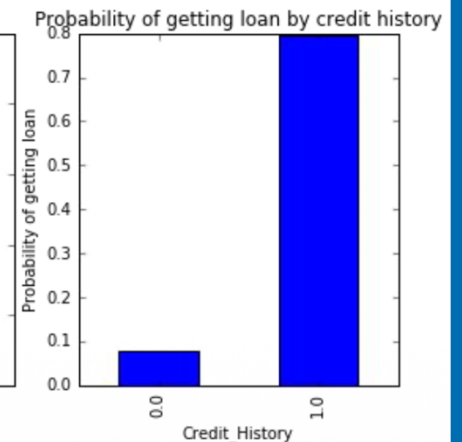
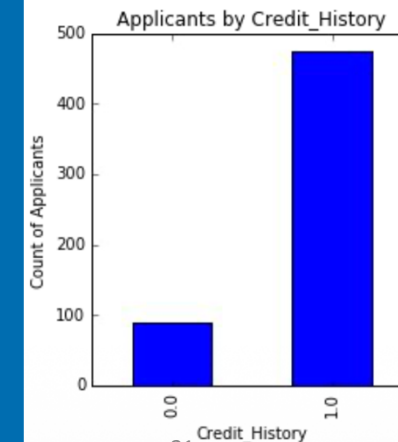
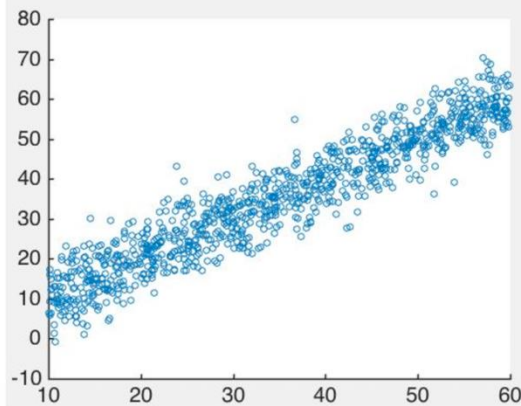
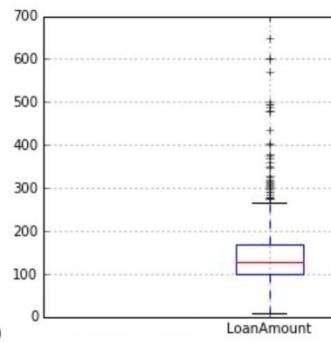
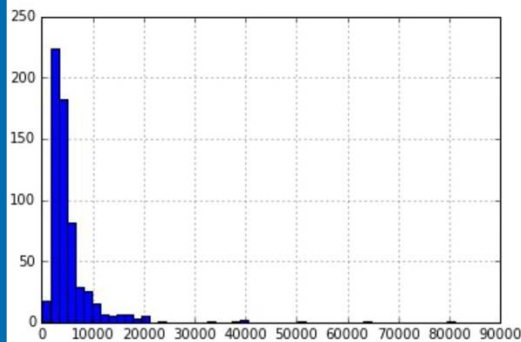
- Inspect rows and variables
- Calculate descriptive statistics
- Visualize distributions and relationships
- Inspect categorical variables

```
In [3]: df.head(10) #Printing first 10 rows of dataset
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount
0	LP001002	Male	No	0	Graduate	No	5849	0	NaN
1	LP001003	Male	Yes	1	Graduate	No	4583	1508	128
2	LP001005	Male	Yes	0	Graduate	Yes	3000	0	66
3	LP001006	Male	No	0	Not a Graduate	No	5849	0	NaN

```
In [4]: df.describe() #Get summary of numerical variables
```

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
count	614.000000	614.000000	592.000000	600.000000	564.000000
mean	5403.459283	1621.245798	146.412162	342.000000	0.842199
std	6109.041673	2926.248369	85.587325	65.120411	0.364878
min	150.000000	0.000000	9.000000	12.000000	0.000000
25%	2677.500000	0.000000	100.000000	12.000000	1.000000
50%	5403.459283	1621.245798	146.412162	342.000000	0.842199
75%	8541.666667	2926.248369	200.000000	360.000000	0.842199
max	16000.000000	16000.000000	4800.000000	600.000000	1.000000



15. Data quality and preparation

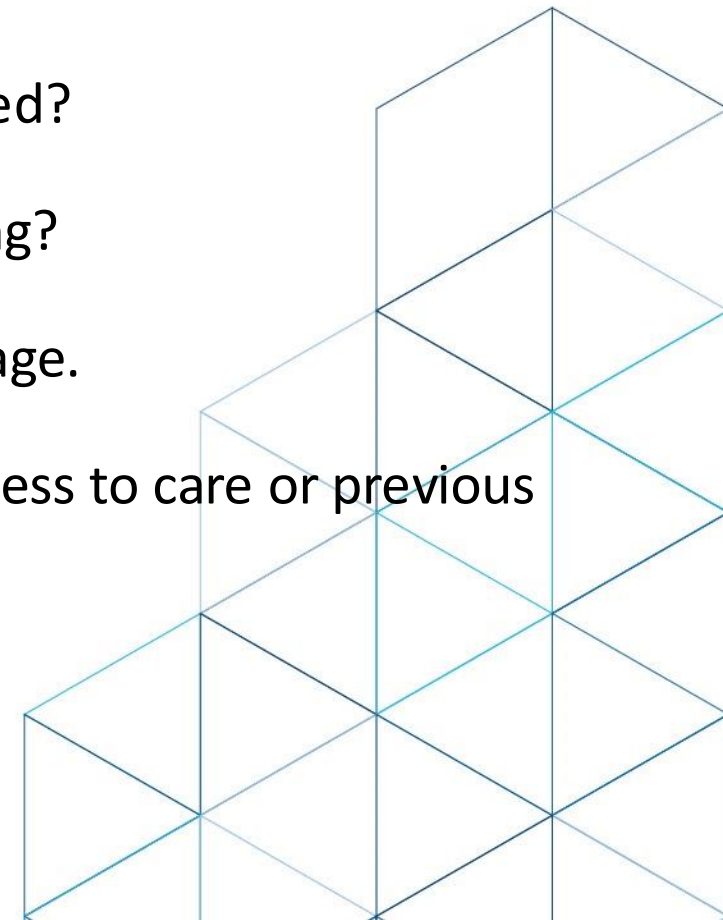
- Check missing values in the dataset:
 - Decide on what to do with them: Deletes entire row/column, imputation (random or based on data distribution), linear interpolation, forward/backward fill...
- Check invalid data:
 - Remove duplicate rows
 - Unify inconsistent format (i.e. units, time zone differences)
 - Correct any other logical inconsistency
- Analyze extreme values:
 - Are they noise? Then we can remove them
 - But what if they are outliers?

Every cleaning decision is an assumption that should be documented

16. A first scientific checkpoint

Before building a model, check if the data *actually* answer our question:

- 1. How were the data collected?**
Were they collected for this purpose—or for something else?
- 2. Who is represented?**
Does the dataset reflect the population where the result will be used?
- 3. What do the variables really measure?**
Is “readmission” defined consistently? Are important factors missing?
- 4. Was the information available at prediction time?**
Using information recorded after discharge would create data leakage.
- 5. What alternative explanations exist?**
An observed relationship may reflect differences in age, health, access to care or previous treatment—not a causal effect.
- 6. Is using the data appropriate?**
Consider privacy, consent, fairness and possible harm.



17. When do we need machine learning?

- **Use machine learning when:**
 - We have enough relevant examples.
 - The desired outcome can be measured.
 - Patterns are too complex or numerous for simple rules.
 - Predictions must generalize to new cases.
 - Performance can be evaluated on unseen data.
 - The output supports a meaningful action.
- **Prefer another approach when:**
 - A summary, visualization or simple rule answers the question.
 - The available data are too limited or unrepresentative.
 - The main question is causal: “What happens if we intervene?”
 - Errors could cause serious harm without adequate human oversight.
 - There is no clear way to use the prediction.

Same topic, different methods

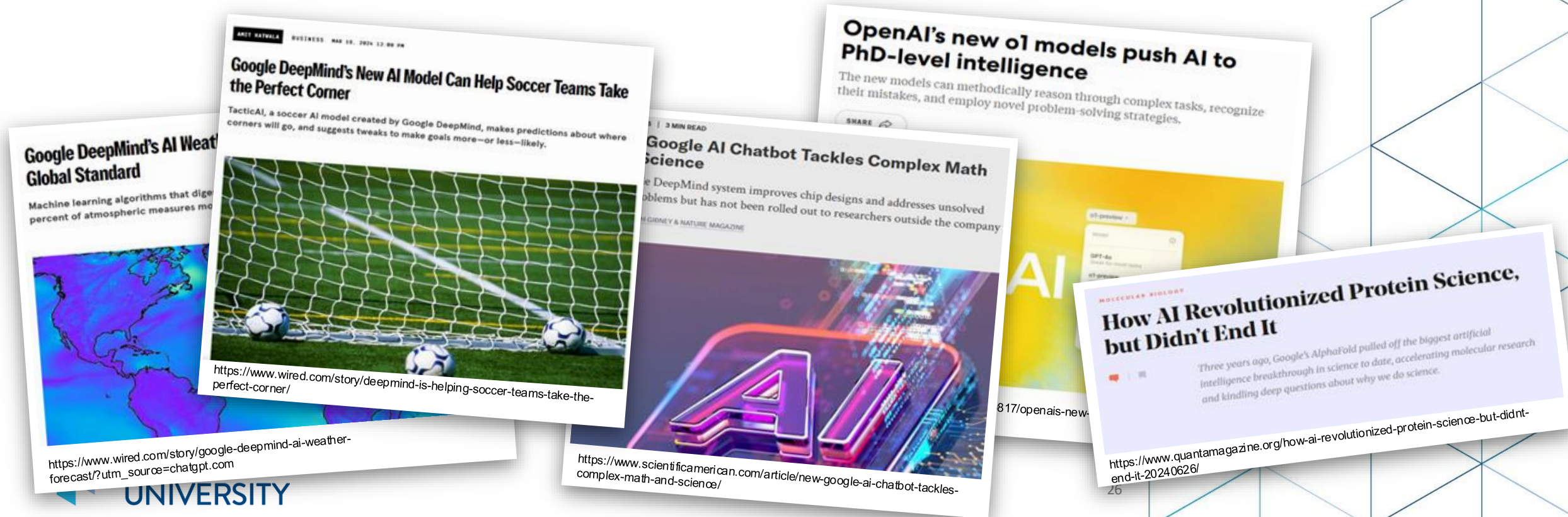
- What was last year's readmission rate?
→ Descriptive analysis
- Which patients may be readmitted? →
Machine learning may help
- Will follow-up visits reduce readmissions? → Causal study or experiment

Part IV – Machine Learning Foundations



18. The power of Machine Learning

- **Machine learning** achieves **state-of-the-art** performance across diverse domains.
- Continuously improves **through data and experience**, not fixed rules.



19. What is machine learning?

- How learning looks like for humans?
 - Studying, observe, experience, practice...

“**Machine Learning**: is a field of study in **artificial intelligence** concerned with the development and study of **statistical algorithms** that can learn from data and **generalize** to unseen data, and thus perform tasks without being explicitly programmed”.

Wikipedia

Traditional Programming

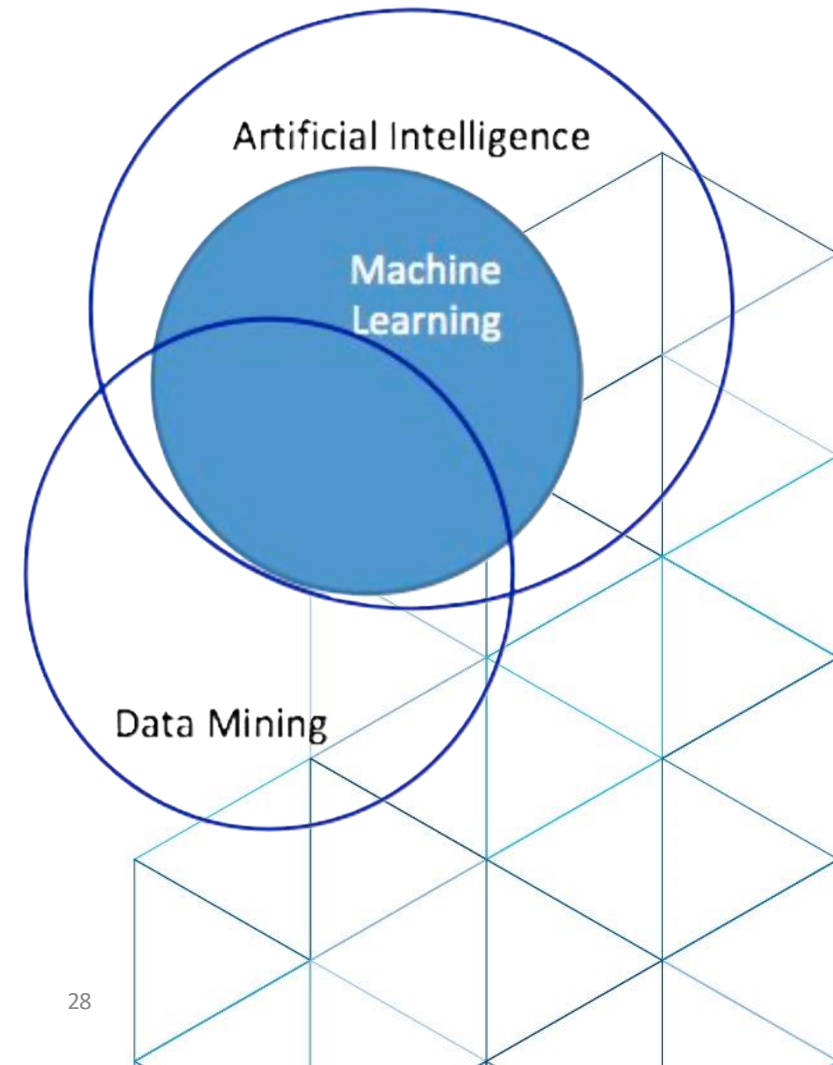


Machine Learning



19. What is machine learning?

- **Artificial Intelligence:** A research field on the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making.
- **Data Mining:** An inter-disciplinary process about extracting and finding patterns in massive data sets involving methods at the intersection of machine learning, statistics, and database systems.
 - It focuses on exploratory data analysis, not in the generalization on unseen data.



20. Traditional programming and machine learning

Traditional programming	Machine learning
- Human-write rules + input data → program → output - The programmer explicitly describes how the system should behave. Spam example: “If the email contains suspicious links, certain phrases and unusual formatting, classify it as spam”	- Training examples + desired outputs → learning algorithm → model - The model learns patterns from previously labelled examples. Spam example: Provide thousands of emails labelled spam or not spam and let the algorithm learn which combinations of characteristics are predictive. New email + trained model → spam probability

- Machine learning still requires human decisions
 - Define the problem and desired outcome
 - Training data and labels
 - Learning method and objective
 - How performance will be evaluated
 - When and how the prediction will be used

21. One task, three elements

Machine Learning is the study of algorithms that

- Improve the task **T**
- with respect to the performance measure **P**
- based on experience **E**

A well-defined learning task is given by **<T, P, E>**

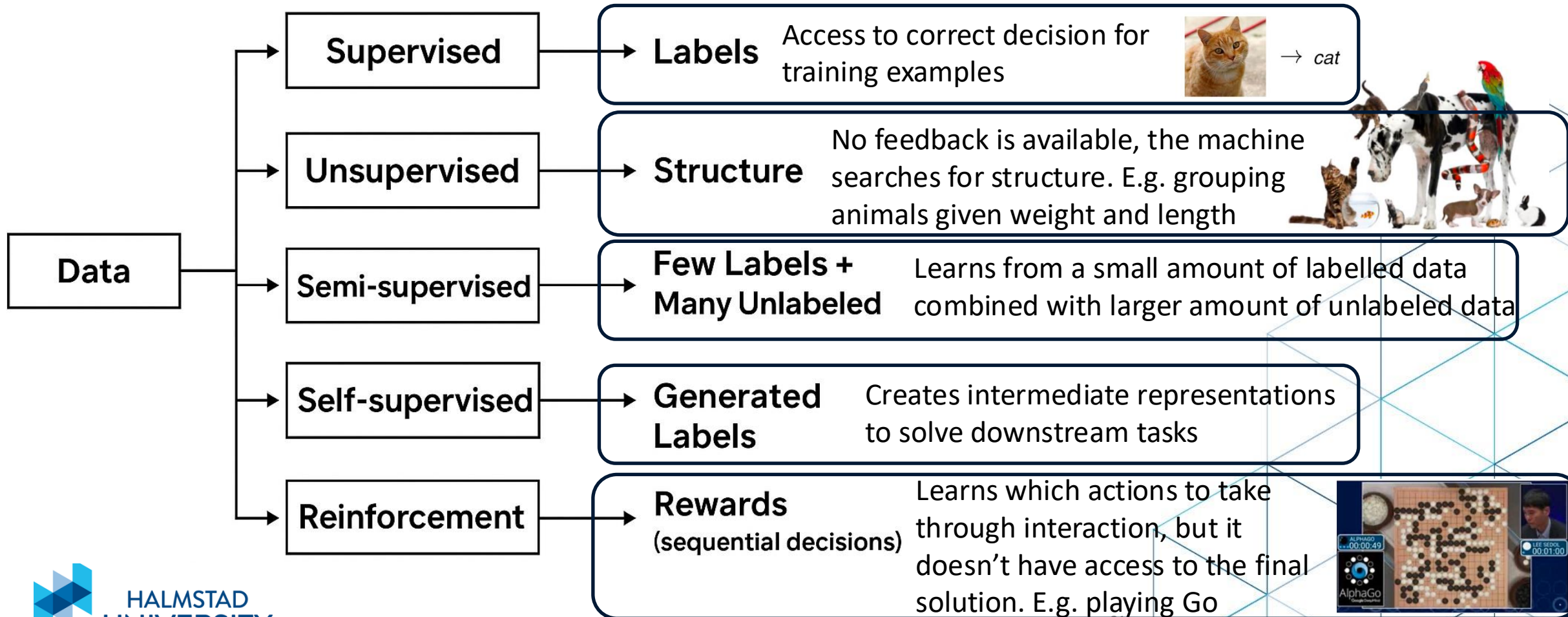
Tom Mitchell, 1998

- What are the learning elements in problems such as:
 - Spam detection?
 - Recognizing hand-written numbers?



22. Main learning paradigms

From data to patterns, guided by labels, tasks, or structure



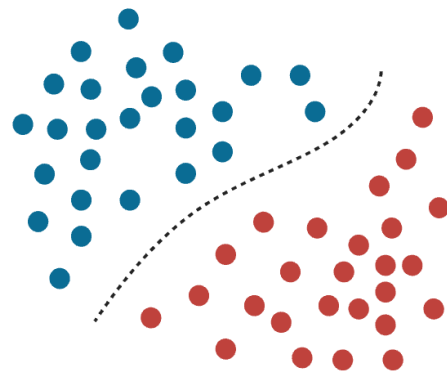
23. Model types: discriminative vs. generative

- **Discriminative models** learn to predict an outcome or separate categories: “Is this email spam?”
- **Generative models** learn patterns in the data distribution and can produce new examples: “Generate an email similar to the training data.”

Discriminative (classic)

Predict a label/class given the features of input data

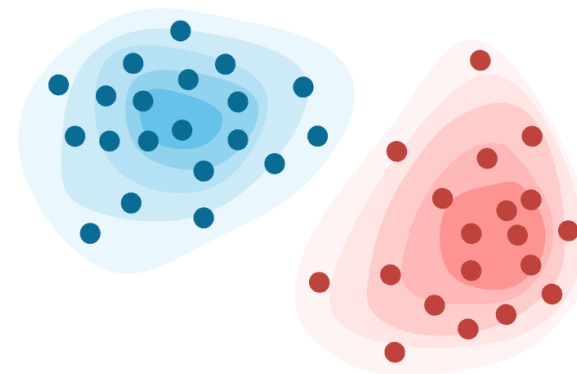
What's learned?: Decision boundary



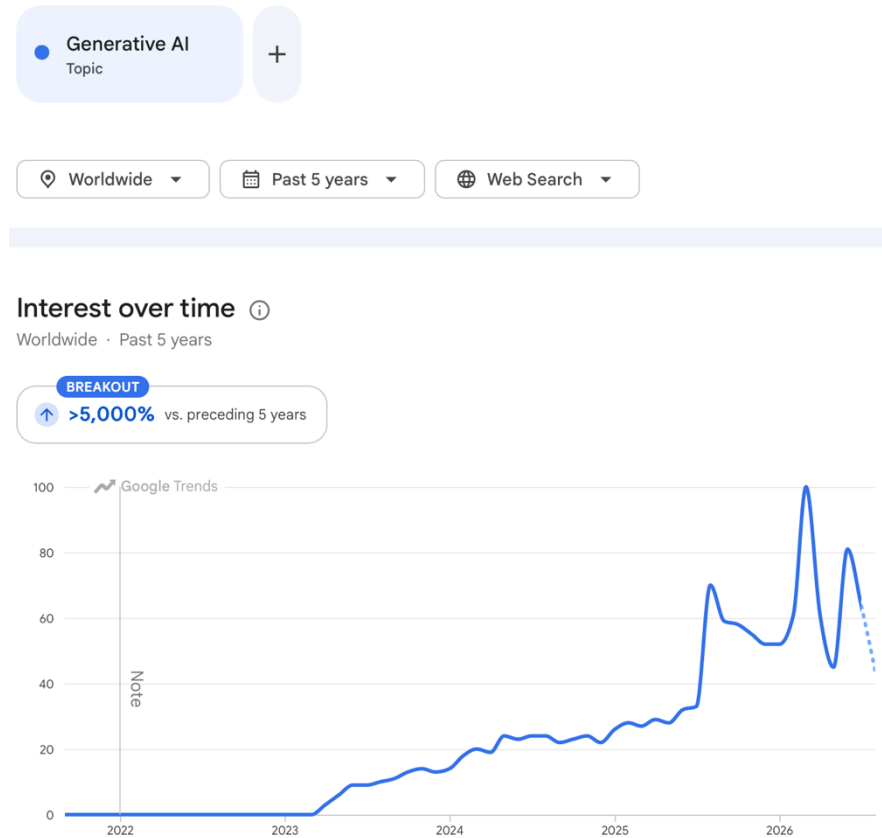
Generative

Abstract underlying patterns in input data in order to generate new content

What's learned?: Probability distributions of the data

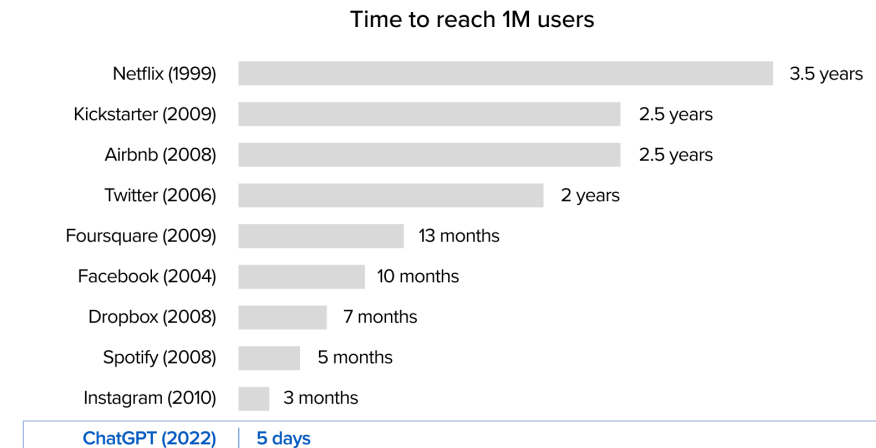


24. Why generative AI matters now?



Google Trends, “Generative AI” topic, worldwide interest over time, as displayed on the slide.

Unprecedented Speed to Adoption



Source: Statista

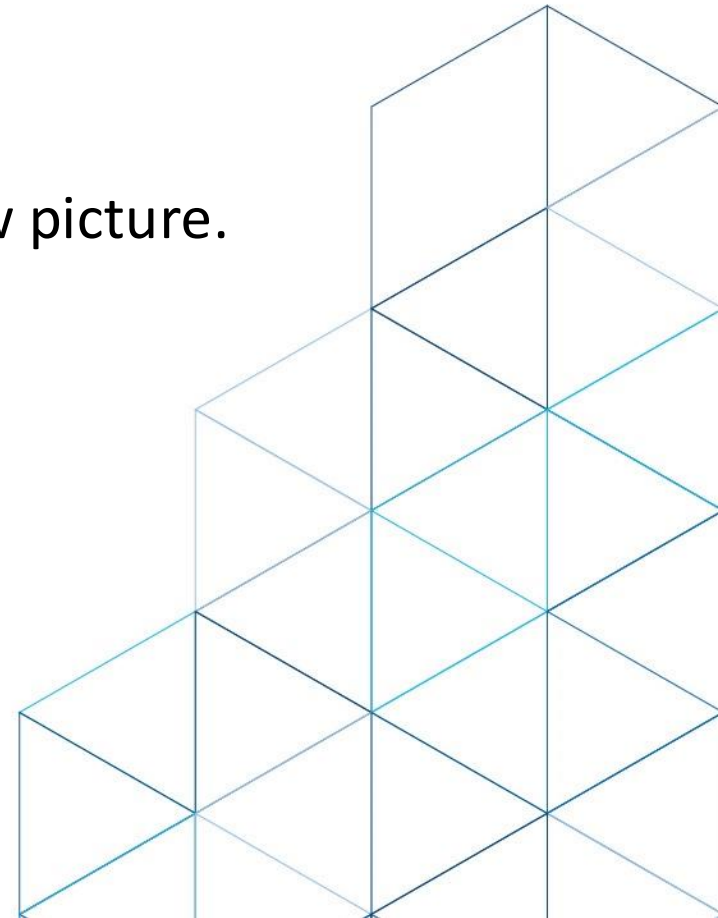
Andreessen Horowitz, “The Economic Case for Generative AI and Foundation Models,” <https://a16z.com/the-economic-case-for-generative-ai-and-foundation-models/>

Part V – Consolidation

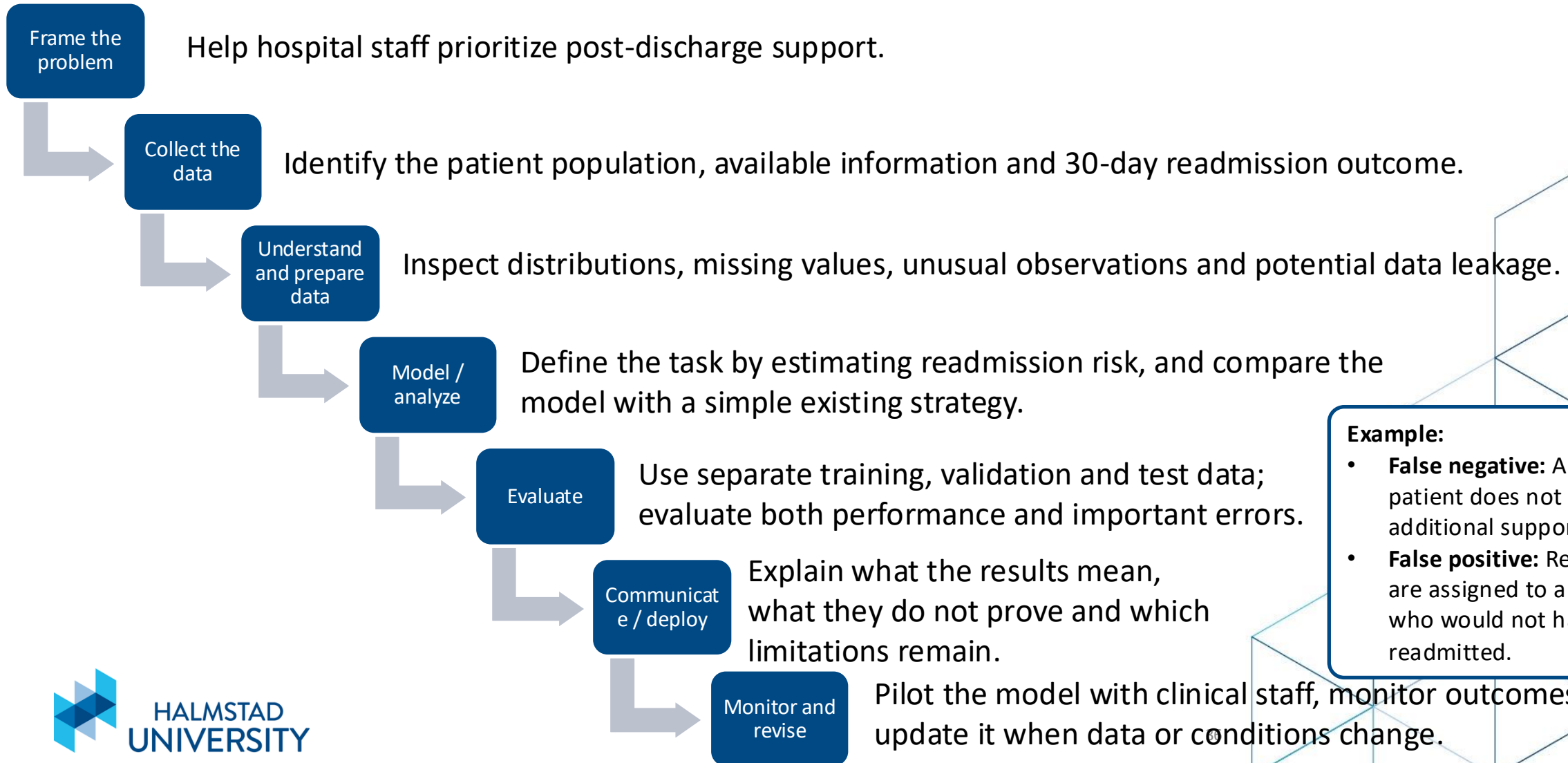


24. Quiz: Which learning type?

1. Analyzing bank data for weird-looking transaction and mark them as fraud.
2. Given someone music choices and a bunch of feature music (tempo, genre, ...), recommend a new song.
3. Music creation by learning existing songs in Spotify.
4. From an album of tagged photos, recognize someone in a new picture.
5. Grouping students based on their learning style.
6. Teaching a robot to grasp a glass by trying many times.



25. From a real problem to an evaluated result



Example:

- **False negative:** A high-risk patient does not receive additional support.
- **False positive:** Resources are assigned to a patient who would not have been readmitted.

26. Your journey through the course

- **Final Seminar:** Present a data science application:
 - Select an interesting application
 - Think about different aspects of such application
 - Connect your application to the content of each lecture
 - Work in groups!

You are going to get a series of assignments that will help you to prepare this task incrementally through the course

26. Your journey through the course

- For this week:
 1. Select an example of a business case using data science.
 2. Explain how can this business benefit from data science.
 3. Which kind of data do they need to acquire for the selected use case?
 4. Search for a dataset on the web that can be used for your use case.
 5. Describe the dataset. Why do you think this dataset is useful for your use case?
 6. Submit the report in Blackboard

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